

BJT Biasing

Faithful Amplification

Inherent Variation of Transistor Parameters and Thermal Runway

Stabilization and Stability Factor

Methods of BJT Biasing

Analysis and Design of Biasing Circuits

Field Effect Transistors

Field Effect transistor (FET):

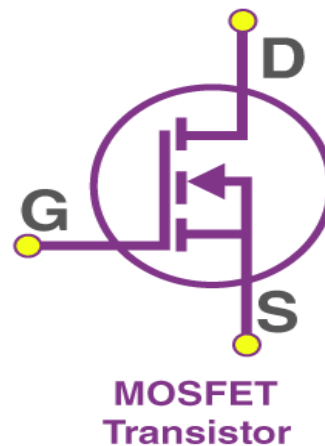
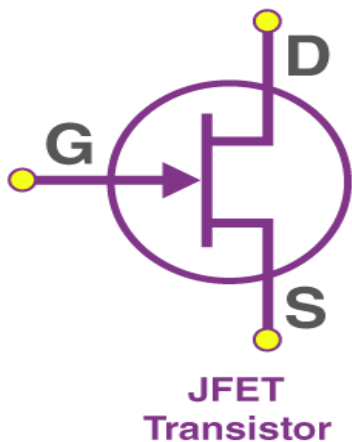
A field-effect transistor (FET) is a type of transistor that uses an electric field to control the current flow through a semiconductor channel. FETs are widely used in electronic circuits due to their high input impedance, low output impedance and high gain.

Classification of FET

There are two types of Field Effect Transistors:

- Junction Field Effect Transistor (JFET)
- Metal oxide semiconductor Field Effect Transistor (MOSFET)

TYPES OF FIELD EFFECT TRANSISTORS



Junction Field Effect Transistor (JFET)

JFET is one of the most basic forms of field-effect transistors. They are three-terminal semiconductor electrical devices that can act as electronically controlled resistors or switches. Unlike a BJT (bipolar junction transistor), a JFET is voltage-controlled and does not require a biasing electrical current. The full-form JFET is Junction-gate Field Effect Transistor. The JFET controls the current flow between the source and the drain terminals by varying the voltage applied to the gate terminal. When a voltage is applied to the gate terminal, it creates an electric field that controls the width of the depletion region in the channel between the source and the drain terminals.

- When the gate-to-source voltage is zero, the depletion region is narrow, and the channel offers low resistance to the current flow between the source and the drain terminals. This is called the “pinch-off” condition, and the JFET is said to be in saturation.
- When the gate-to-source voltage is decreased in a JFET (junction field-effect transistor), the depletion region around the channel between the source and drain increases in width. This reduces the effective channel width, which in turn, decreases the conductivity of the channel. As a result, the drain current decreases, and the JFET becomes more resistive.
- When the gate-to-source voltage increases in a JFET (junction field-effect transistor), the depletion region around the channel between the source and drain decreases in width. This increases the effective channel width, which in turn, increases the conductivity of the channel. As a result, the drain current also increases, and the JFET becomes less resistive.

Metal-Oxide Semiconductor Field-Effect Transistor (MOSFET)

The MOSFET is another type of field-effect transistor (FET) that can switch electronic signals or amplify them. It is made of an insulated gate that controls the device’s conductivity per the voltage applied. This nature makes it useful for switching electronic signals or amplifying by adjusting its conductivity with changing voltage levels.

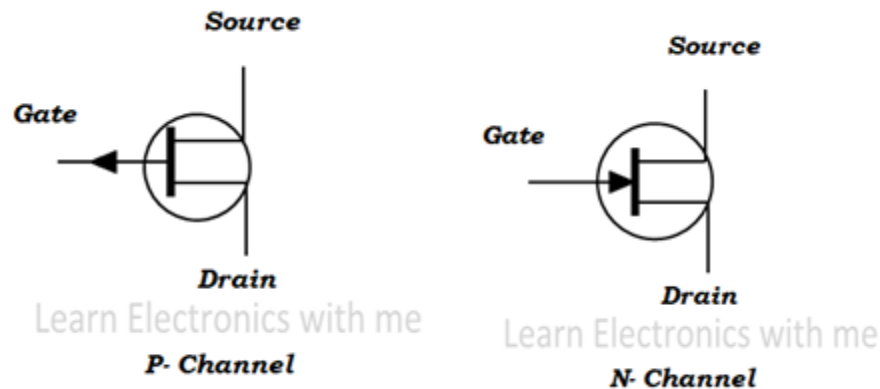
The crucial advantage of MOSFET is that they require almost no input current to regulate the load current relative to bipolar transistors (BJTs or bipolar junction transistors). There are two types of MOSFETs: enhancement mode and depletion mode. In the case of an enhancement mode MOSFET, the exerted voltage at the gate increases the conductivity. In depletion mode transistors, the voltage exerted at the gate decreases the conductivity.

Construction, Operation and Characteristics of JFET

JFET is one of the types of FET. It is voltage-controlled device. JFET is used as a voltage-controlled resistor or switch or as an amplifier. There are two types of JFET namely N-Channel JFET and P-Channel JFET.

In N channel the generation of the current is due to the movement of electrons, in P channel the generation of current is due to the movement of holes. Since the movement of electrons is faster than the movement of holes, N- Channel JFET is preferred more than the P-Channel JFET.

Symbol of N channel and P channel JFET:



Symbol of N channel and P channel JFET

JFET has three terminals Drain, Source and Gate equivalent to Collector, Base and Emitter in BJT. BJT have PN Junctions but JFET have channels made of either P type or N type. In P-Channel JFET arrow points outside and in N-Channel JFET arrow points inside.

Three terminals of JFET:

Source:

The majority carriers enter the Field Effect transistor through the source terminal.

Gate:

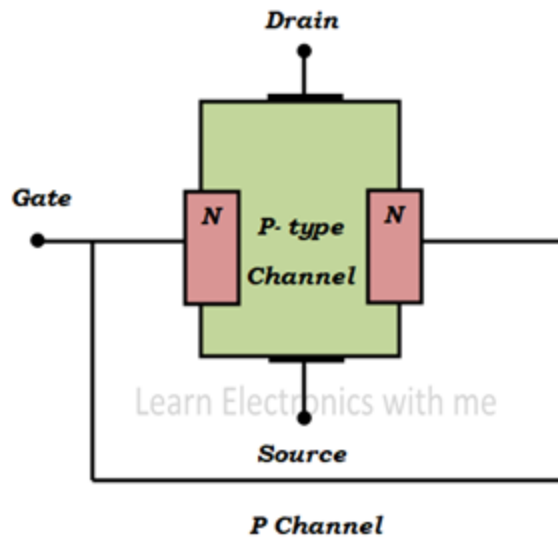
By controlling the Gate voltage, the flow of majority carriers from Source to drain can be controlled.

Drain:

The majority charge carriers leave the channel through drain. This current is designated by the source gate voltage V_{GS}

Channel:

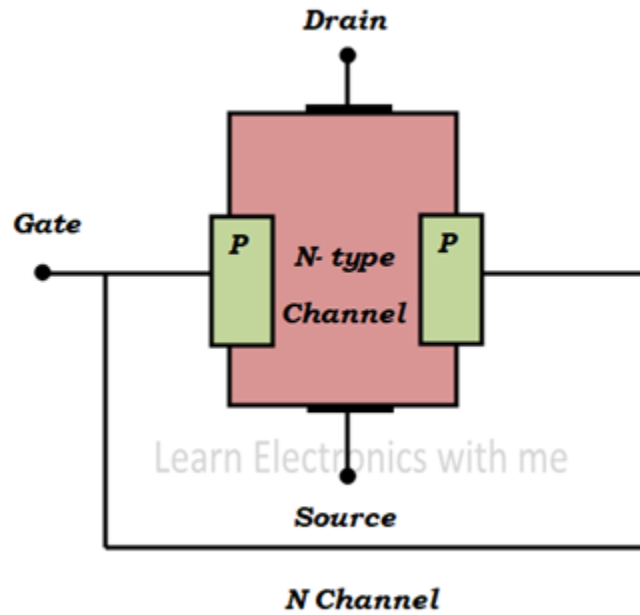
The region between source and drain and also between the gate region is called the channel.

Construction of N channel and P channel JFET:

Construction of P channel

As it is shown in the figure in P Channel JFET in the P type substrate drain and source terminals are taken by creating ohmic contacts.

N type material is used at the either side of the channel and the gate terminals are taken from the N type materials.

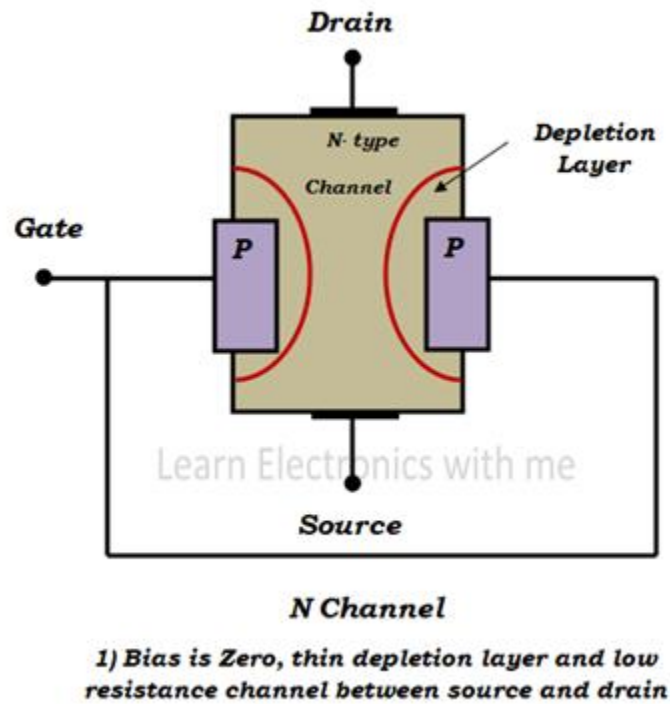


Construction of N channel

In N Channel JFET in the N type substrate drain and source terminals are taken by creating ohmic contacts and P type material is used at the either side of the channel and the gate terminals are taken from the P type materials.

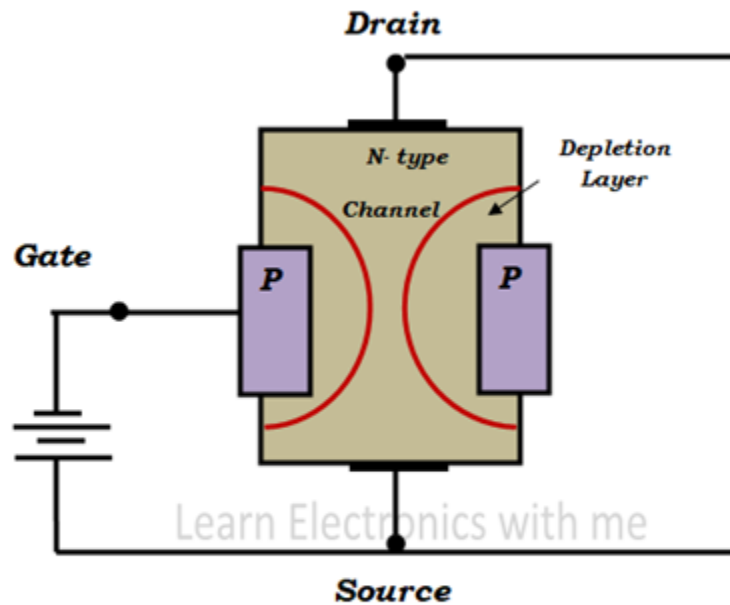
The drain and source terminal are taken from the both the sides of the channel. A small voltage at the Gate terminal controls the current flow between source and drain.

Working of N channel and P channel JFET:



Working of N channel when Bias is zero

The Gate to source voltage controls the flow of current from source to drain. When no gate to source voltage is applied or when bias is zero the resistance in the channel is low. So, when a small drain to source voltage is applied it causes the current to flow from source to drain.

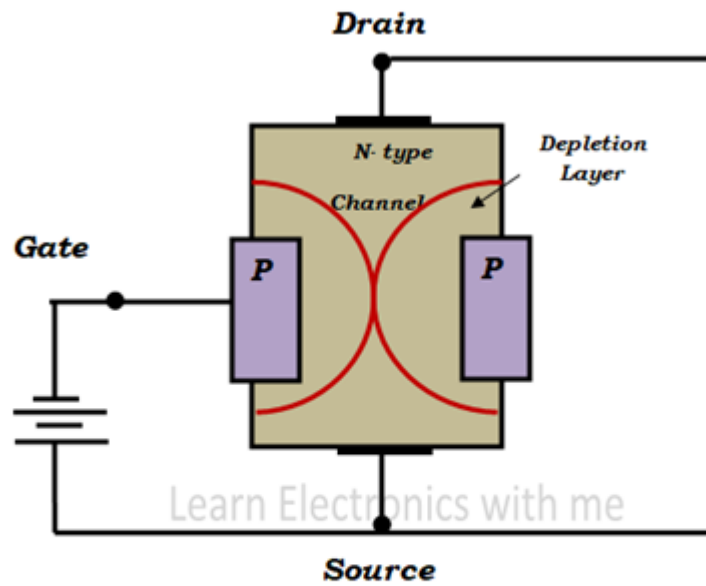


N Channel

2) Moderate gate to source reverse bias, narrow channel

Working of N channel when moderate bias is applied

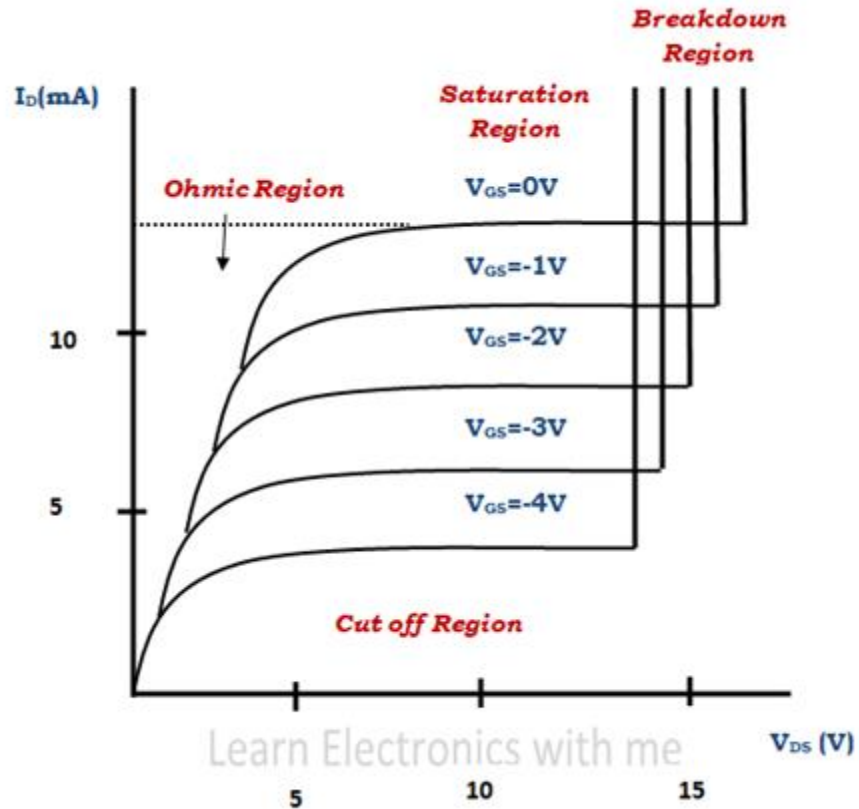
When the gate terminal is reverse biased or when negative gate source voltage is applied the size of the depletion layer increases and the resistance in the channel increases. Thus, it controls the flow of current in the channel.



Working of N channel when bias greater than Pinch off is applied

When the reverse bias at the gate terminal is increased the depletion, region increases and they meet at a point of time. This is called Pinch off mode. At this point the source to drain current is zero, but some reverse saturation current exists. The gate to source voltage at which Pinch off occurs is called Pinch off voltage.

V-I Characteristics of JFET:



Ohmic Region:

It is the region when gate source voltage is zero $V_{GS}=0$

Saturation Region:

In this region JFET is ON and active.

Cut Off Region:

JFET is OFF in this region and there is no drain current flowing.

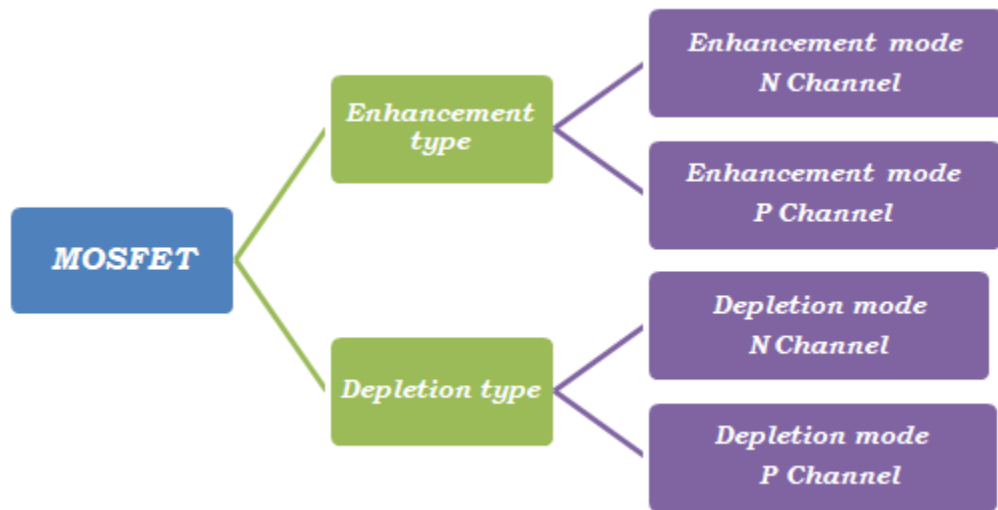
Breakdown Region:

When the drain voltage exceeds the maximum necessary voltage, current flows from drain to source without any resistance and the device breakdown and the JFET enters the breakdown region.

Operation and Characteristics Curves of MOSFET

MOSFET stands for Metal Oxide Semiconductor Field Effect Transistor. It is a type of Field Effect Transistor and it is voltage-controlled device. It is also called as Insulated Gate Field Effect Transistor (IGFET). It is used for switching or amplifying electronic signals in the electronic devices. It is the most commonly used transistor and it can be used in both analog and digital circuits.

Types of MOSFET

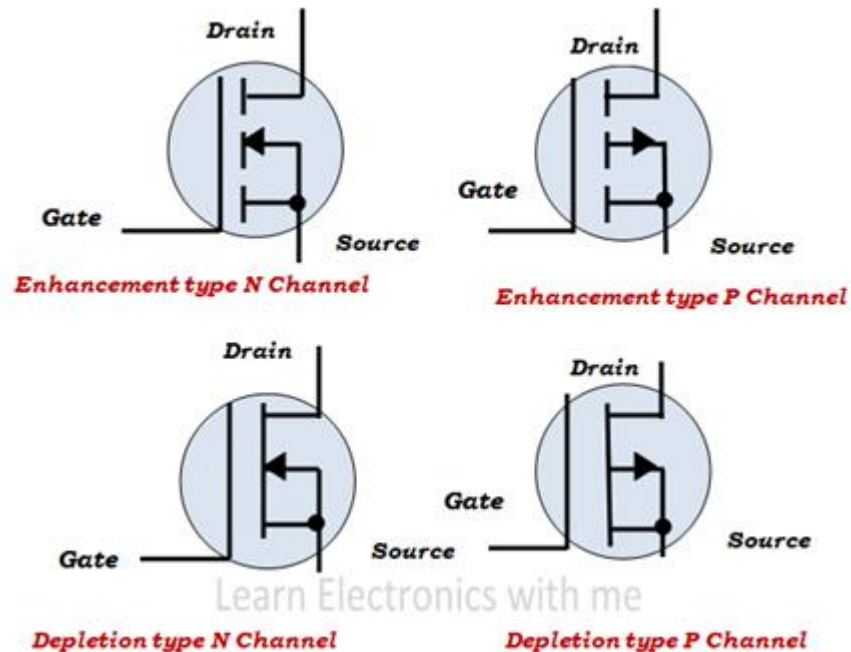


Types

of

MOSFET

Symbol of MOSFET:

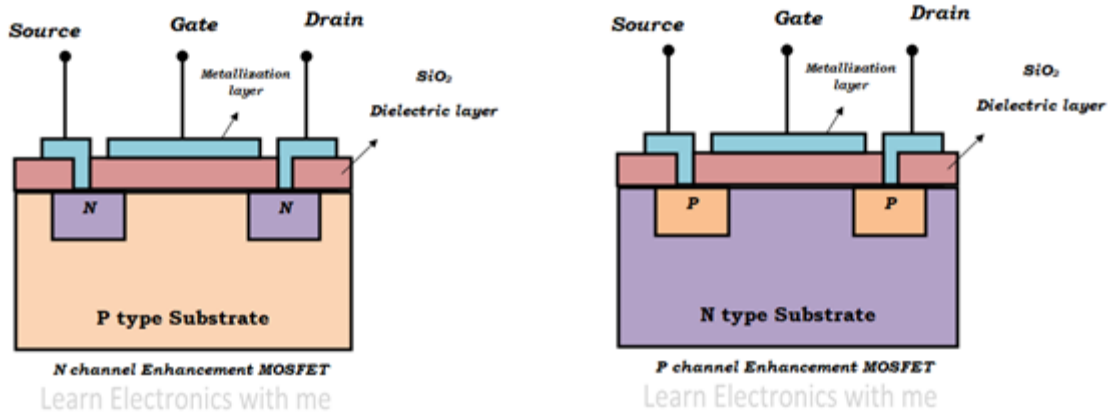


Symbols of MOSFET

In the Enhancement MOSFET the source and the drain are not connected physically so in the symbol lines are broken and in the Depletion mode line is continuous. In the N type the arrow points inside and in the P type arrow points outside.

Construction of MOSFET

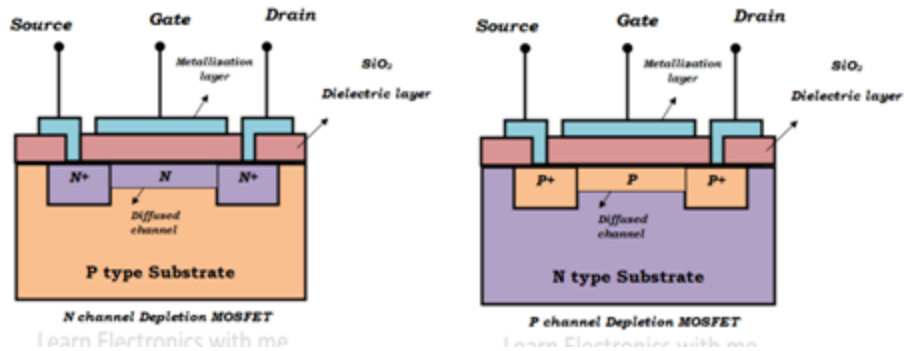
The metallic gate terminal in the MOSFET is insulated from the semiconductor layer by a SiO_2 layer or dielectric layer. The MOSFET consists of three terminals, they are source(S), Gate (G), Drain (D) and the body which is called as substrate. The substrate is connected to the source internally.



Construction of N channel and P channel Enhancement MOSFET

In N channel Enhancement MOSFET the source and drain are of N type semiconductor which is heavily doped and the Substrate is of P type semiconductor. Majority charge carriers are electrons. The source and drain terminals are physically separated in Enhancement mode.

In P channel Enhancement MOSFET the source and drain are of P type semiconductor which is heavily doped and the Substrate is of N type semiconductor. Majority charge carriers are holes.

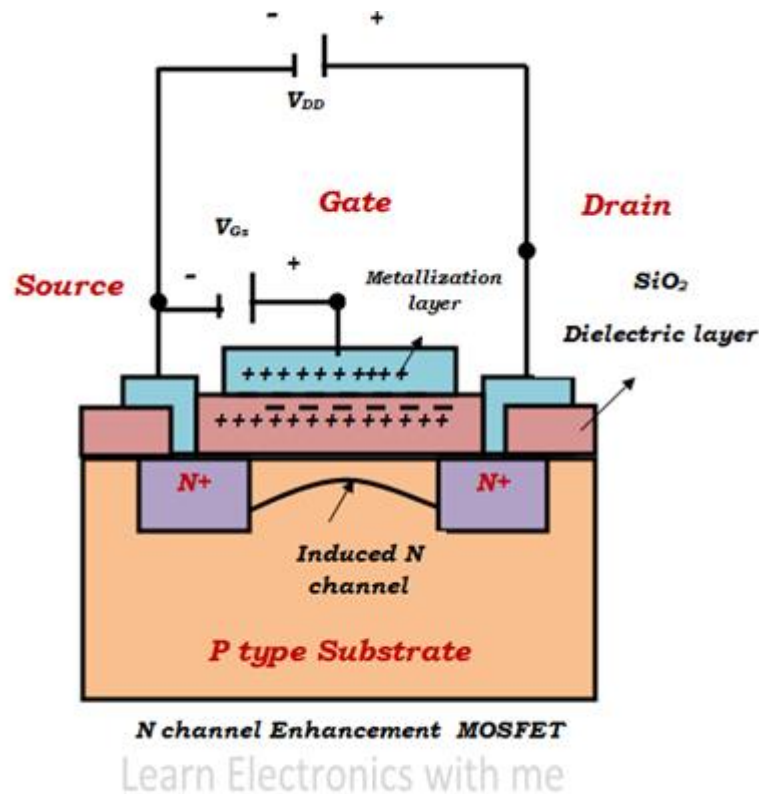


Construction of N channel and P channel Depletion MOSFET

In the N Channel depletion MOSFET a small strip of semiconductor of N type connects the source and drain. The source and drain are of N type semiconductor and the Substrate is of P type semiconductor. Majority charge carriers are electrons. The source and drain are heavily doped.

In the P Channel depletion MOSFET a small strip of semiconductor of P type connects the source and drain. The source and drain are of P type semiconductor and the Substrate is of N type semiconductor. Majority charge carriers are holes.

Working of Enhancement MOSFET:



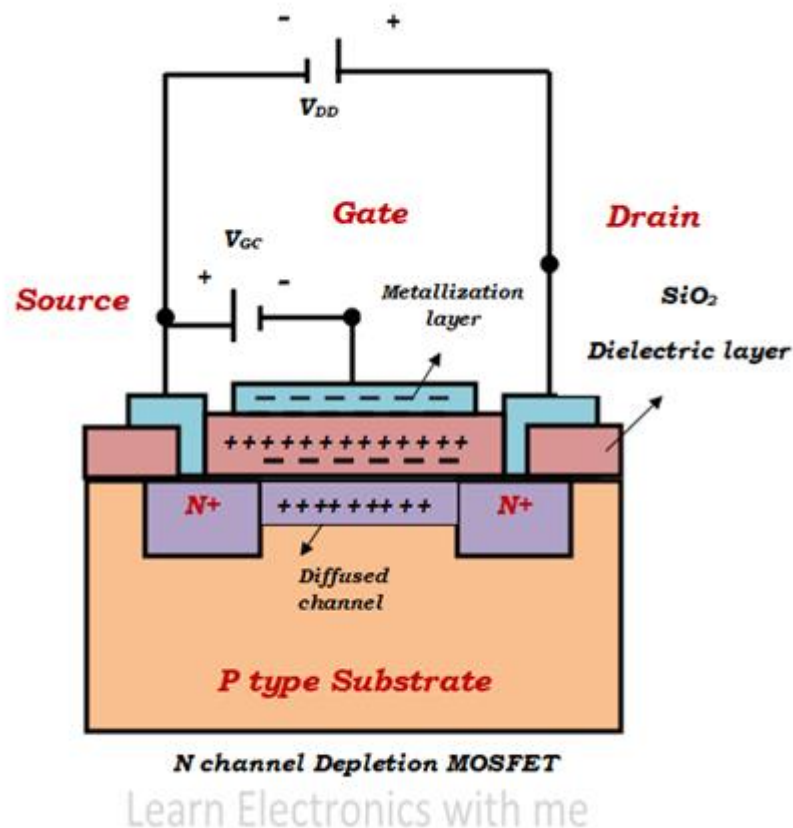
Working of N channel Enhancement MOSFET

In the enhancement mode the applied Gate voltage is always positive. When it crosses the threshold voltage it turns ON. The current is generated due to the movement of majority carriers. In N channel majority carriers are electrons and in P channel majority carriers are holes. The source is connected to the negative terminal. When the electrons move from source to drain the positive charges formed below the dielectric because of the repulsive force from gate combine with each other.

When the applied gate voltage is increased the number of majority carriers becomes more than the minority carriers below the dielectric medium. So, the majority carriers overcome the

recombination of holes and electrons and the majority carrier move from source to drain in the channel, which forms the current. Thus, the gate voltage controls the concentration of the majority carriers which is responsible for the formation of the channel.

Working of Depletion MOSFET:



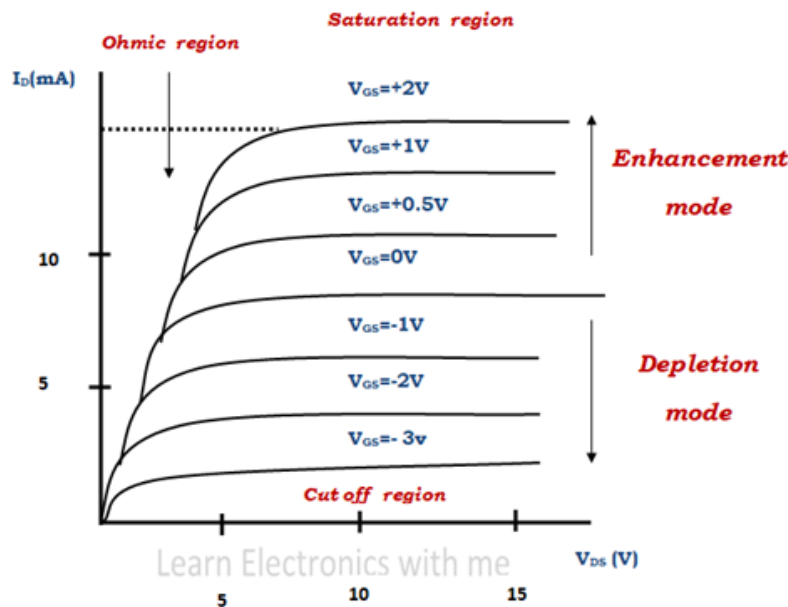
Working of N channel Depletion MOSFET

The depletion MOSFET is ON by default. The source and drain terminals are physically connected. When the gate terminal is connected to the negative terminal and source to the positive terminal, the electrons get repelled below the dielectric layer. The positive charged carrier from the source gets combined with the majority carrier the electrons in the N type and thus depletion layer is formed and the channel resistance increases and the current flow decreases. Thus, the

increase in gate voltage decreases the drain current. They are inversely proportional. When the negative voltage is further increased it reaches the pinch off mode.

When the gate is connected to the positive terminal and the source terminal it operates in the enhancement mode.

V-I Characteristics of MOSFET:



V-I Characteristics of MOSFET

Cut off region:

No current flows through it and the MOSFET is off.

Ohmic region:

Drain current increases when the drain source voltage increases. Used as amplifier in this region.

Saturation region:

Drain current is constant for drain source voltage. Used as switch in this region. This occurs when the drain source voltage reaches pinch off voltage.

Depletion mode:

The MOSFET is ON by default. When negative voltage is applied to the gate terminal it operates in the depletion mode and when positive voltage is applied, it operates in the enhancement mode.

Enhancement mode:

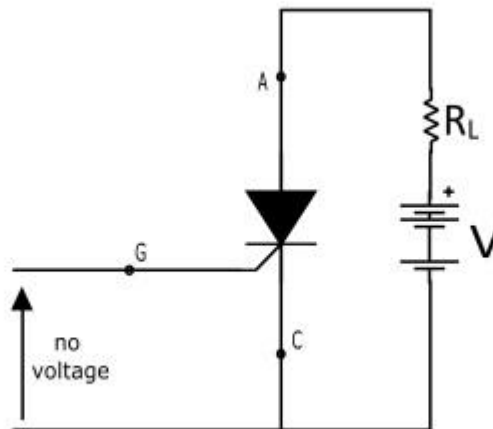
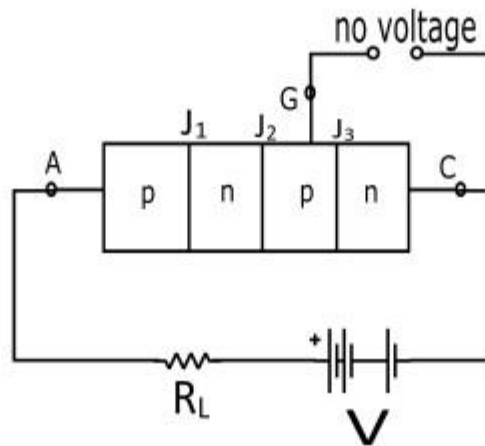
When positive voltage is applied to the gate terminal, it starts conducting and the current starts to flow.

Transfer Characteristics and Shockley's Equation**DC Biasing of JFET****Characteristics and Applications of Industrial Electronics Devices:****SCR (Silicon Controlled Rectifier)**

A silicon controlled rectifier or semiconductor controlled rectifier is a four-layer solid-state current-controlling device

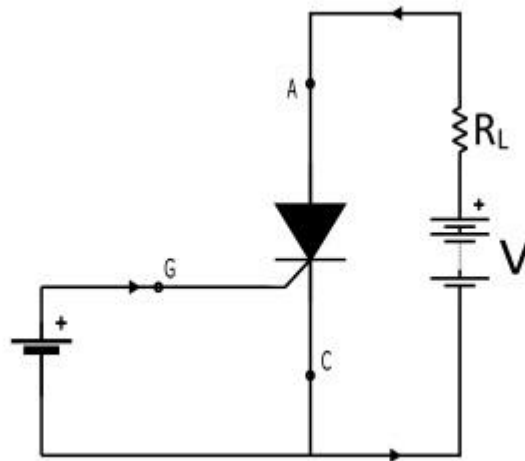
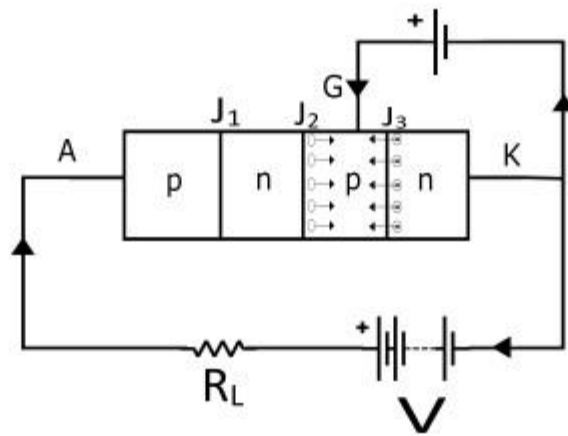
Working Principle of SCR**When Gate is Open Circuited**

When no voltage applied to gate terminal, junction J₂, is reverse biased and the junctions J₁ and J₃ are forward biased. Since one of the three junctions is reverse biased so there is no current can flow through the load, hence the SCR is OFF. However, if the applied voltage is gradually increased, a stage is reached, when reverse biased junction (J₂) breaks down. The SCR now, starts conducting and become ON. The value of applied voltage at which the reverse biased junction breaks down and the SCR becomes ON is known as Breakover Voltage.



When Gate is Positive with Respect to Cathode

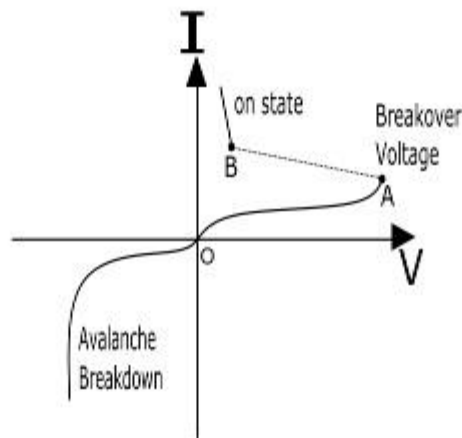
The SCR can be turned ON at smaller applied voltage by the application of a small positive voltage at the gate terminal. When gate voltage is applied the junction J_3 is forward biased and junction J_2 is reverse biased. Thus, the electrons from n – type layer starts moving across the junction J_3 toward p –type material and the holes from p –type material towards the n – type material. Due to the movement of holes and electrons across the junction J_3 the gate current starts flowing. Because of gate current the anode current increases. The increased anode current makes the more electrons available at the junction J_2 . As a result of this process, in a small time, the junction J_2 breaks down and the SCR is turn ON.



Even if the voltage at the gate terminal is removed, the SCR the anode current does not decrease. The SCR can only be turned off by reducing the applied voltage to zero.

I-V Characteristics of SCR

It is the curve plotted between the anode – cathode voltage (V) and anode current (I) of the SCR at constant gate current.



- **Forward Characteristics** – When the anode is made positive with respect to the cathode, then the curve between V and I is called as forward characteristics of the SCR. If the supply voltage is increased from zero, a point is reached (Point A, the voltage is called breakover voltage) when the SCR starts conducting. Under this condition, the voltage across SCR decreases suddenly (shown by dotted line in the curve) and the most of the supply voltage appears across the load.
- **Reverse Characteristics** – When anode is made negative with respect to the cathode, the curve plotted between V and I is called as reverse characteristics. If the reverse voltage is increased gradually, at first the anode current remains small (called leakage current) and at some reverse voltage, the avalanche breakdown occurs and the SCR starts conducting in the reverse direction (Shown by the curve in third quadrant). The maximum reverse voltage at which the SCR starts conducting in the reverse direction is called as Reverse Breakdown Voltage.

Applications of SCR

The SCR can be used in following application –

- Power Switching Circuit
- Controlled Rectifier
- AC power control circuits
- Speed control of DC shunt motor
- SCR Crowbar
- Computer logic circuits
- Timing Circuits

- Inverters
- Battery Charging Regulators
- Temperature control systems

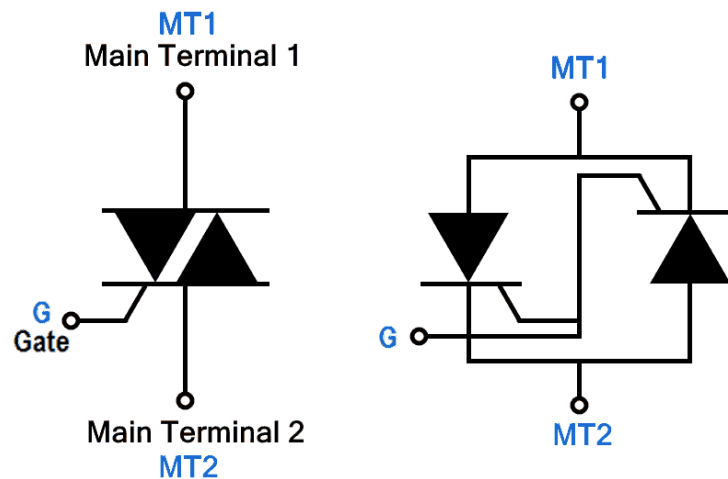
TRIAC

TRIAC is an acronym that stands for “Triode for Alternating Current”. Triode means a three-terminal device while AC means that it is used for switching Alternating current. It is a three-terminal bi-directional switch that conducts in both directions. It is made from the combination of two SCRs in anti-parallel with their gates joined together.

Working principle of TRIAC

The operation of TRIAC resembles thyristor. When the voltage is applied, it will not conduct unless the voltage does not exceed the break over voltage V_{BO} or a gate pulse is applied.

As we know that TRIAC can conduct for both polarities of applied voltage and it can be triggered by both polarities of gate voltage for either direction. Therefore, the TRIAC can operate in 4 modes.



TRIAC Symbol

Fig-Working Diagram of TRIAC

The following voltages are taken with respect to the MT2 terminal such as MT1 voltage with respect to MT2 and Gate voltage with respect to MT2.

Mode 1: $MT1 = +ve, Gate = +ve$

In this mode, the applied voltage at MT1 is positive with respect to MT2. By applying a positive gate pulse, the TRIAC will trigger into **forward conduction** and the current will flow from MT1 to MT2.

Mode 2: $MT1 = +ve, Gate = -ve$

In this mode, the applied voltage is the same i.e. MT1 is positive with respect to MT2. But the gate pulse is negative. Since the gate is connected with the N region of TRIAC, it will trigger it into **forward conduction** while the current direction will remain the same.

Mode 3: $MT1 = -ve, Gate = +ve$

In this mode, the applied voltage polarities are swapped i.e. MT1 is negative with respect to MT2. But the gate pulse is positive. The gate pulse will trigger TRIAC into **reverse conduction** from MT2 to MT1.

Mode 4: $MT1 = -ve, Gate = -ve$

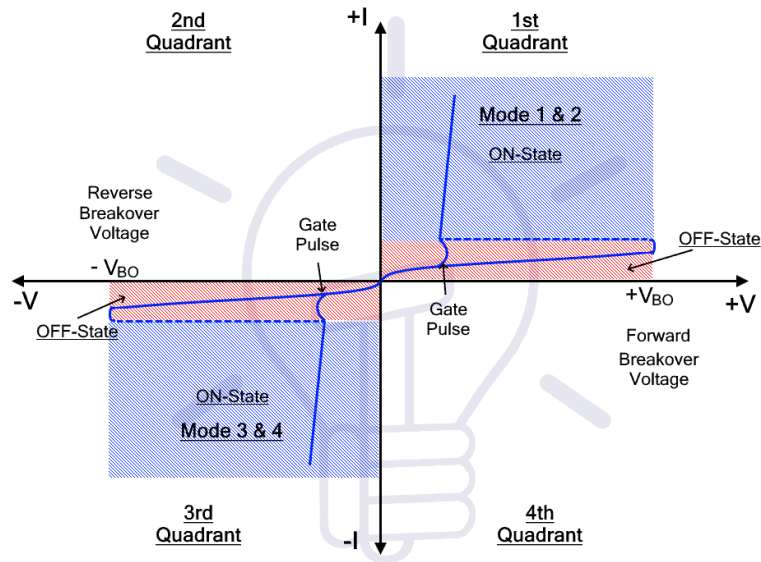
In this mode, both the applied voltage and the gate voltage are negative. The negative gate pulse triggers the TRIAC into **reverse conduction**.

Mode 1 and mode 2 represents an operation in the 1st quadrant where the current and voltage are positive while mode 3 and mode 4 represent operation in the 3rd quadrant where the voltage and current both are negative.

While the gate pulse can trigger TRIAC in either direction, it is best to use a positive gate pulse for operation in the 1st quadrant and negative gate pulse for operation in the 3rd quadrant due to their increased sensitivity. Mode 2 and 3 require more gate current than mode 1 and 4 to trigger TRIAC.

V-I Characteristic of TRIAC

The following characteristic curve shows the relationship between the applied voltage and the current flowing through the TRIAC. It operates in only the 1st and 3rd quadrants. Its operation is the same as SCR but it can also operate in the 3rd quadrant.



V-I Characteristics of TRIAC

The current I increase when either voltage V exceeds the break over voltage V_{BO} or if a gate pulse is applied. As soon as, the device switches into ON-state, the voltage reduces to ON-state voltage and the current exceeds. It will remain in ON-state until the current reduces below holding current I_H .

TRIAC is the combination of two SCR in a single package, therefore, it also has the same electrical characteristics as individual SCR in each direction such as break down voltage, trigger voltage, holding current.

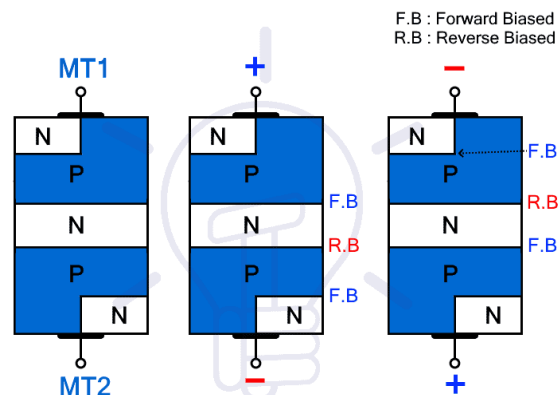
DIAC

DIAC is an acronym that stands for “Diode for Alternating Current”. It is a two-terminal bidirectional switch that conducts in both directions when the applied voltage exceeds its break over voltage. It cannot amplify or offer controlled switching.

Working Principle of DIAC:

The DIAC can conduct current in both directions unless the applied voltage falls below the break over voltage.

Suppose the applied voltage at MT1 is positive with respect to MT2, the junctions at the ends become forwards biased and the middle junction becomes reverse biased. At this moment, the applied Voltage $V < V_{BO}$, so the middle junction remains reverse biased and does not allow current flow. The device remains in off-state.



Working of DIAC

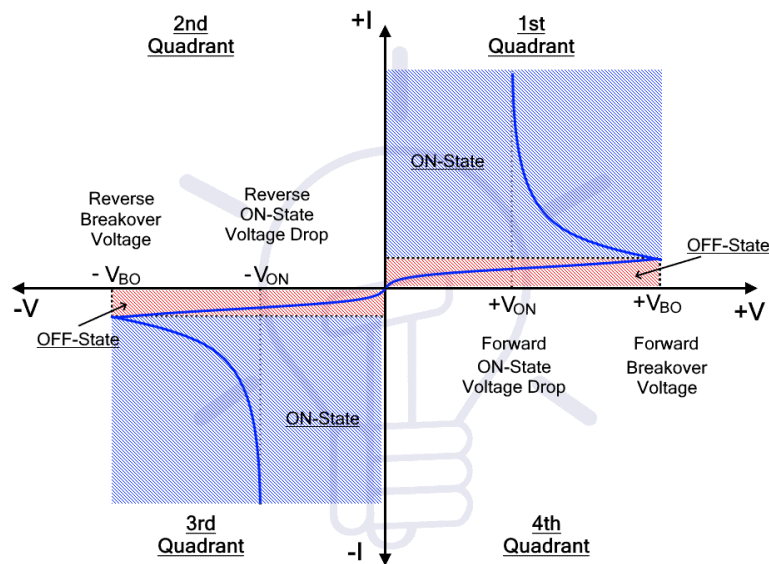
In order to trigger the DIAC into conduction, the applied voltage V must exceed break over voltage V_{BO} . When it happens, avalanche break down occurs at the reverse bias junction and the current starts to flow through it. The DIAC is triggered into conduction and the voltage across it reduces to ON-state voltage drop.

Similarly, if the voltage polarities are swapped, the same process will repeat except the current will flow in reverse direction. There is no difference in operation whatsoever if the polarities are swapped. It has symmetrical switching characteristics for both voltage polarities i.e. its forward break over voltage is equal to reverse break over voltage.

The DIAC conducts current unless the current falls below the holding current limit. As soon as it falls below the said limit, the device switches into off-state.

V-I Characteristics of DIAC

The following characteristic curve shows the relation between the main voltage and main current of a DIAC. As you can see, the DIAC only operate in 1st and 3rd quadrant. It is a symmetrical device; therefore, the graph is symmetrical in both quadrants forming the shape of letter “Z”.



V-I Characteristics of DIAC

In 1st quadrant, the voltage and current are positive. As you can see, when the voltage is below the break over voltage V_{BO} , the DIAC blocks the current except for the leakage current. The device remains in OFF-state. Once the voltage increases, the DIAC triggers into ON-state and the current rises. The voltage across the device starts to reduce to steady ON-state voltage.

The device operates similarly in the 3rd quadrant. The only difference between the 1st and 3rd quadrant is that the voltage and current are reverse. That is, it.

UJT (Unijunction Transistor)

A Unijunction Transistor (UJT) is a three-terminal **semiconductor device**. The main characteristics of UJT is when it is triggered, the emitter current increases re-generatively until it is limited by

emitter power supply. Due to this characteristic feature, it is used in applications like switching pulse generator, saw-tooth wave generator etc.

Working Principle of Unijunction Transistor

Feedback Techniques and Op-amps:

Negative and Positive Feedback

- **Positive feedback**

The feedback signal is in phase with the input signal, increasing gain but also increasing distortion and instability. Positive feedback is often used in oscillators and comparators to provide hysteresis.

- **Negative feedback**

The feedback signal is out of phase by 180° with respect to the input signal, reducing gain but also reducing distortion and overall noise. Negative feedback is often used in amplifiers to increase stability and decrease output impedance.

Characteristics and Gain with Negative Voltage and Current Feedback

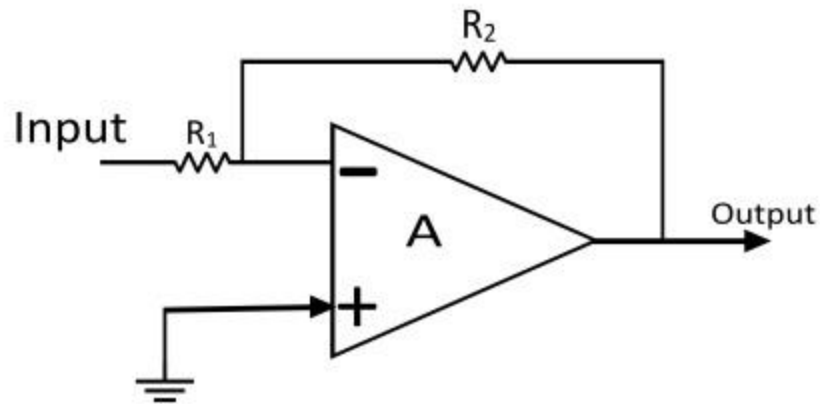
Emitter Follower

Operational Amplifier

An operational amplifier is a three-terminal device consisting of two high impedance input terminals, one is called the inverting input denoted by a negative sign and the other is the non-inverting input denoted with a positive sign. The third terminal is the output of the Op-Amp.

Inverting Operational Amplifier

In the inverting operational amplifier circuit, the signal is applied at the inverting input and the non-inverting input is connected to the ground. In this type of amplifier, the output is 180° out of phase to the input, i.e. when positive signal is applied to circuit, the output of the circuit will be negative.



Applying KCL at inverting node of Op-Amp

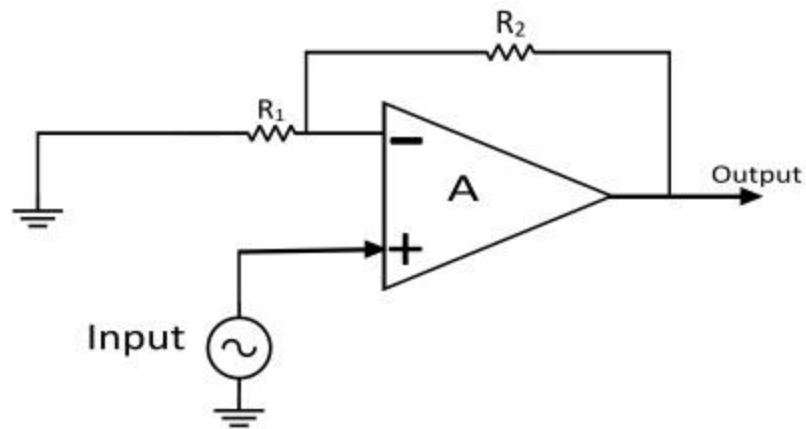
$$\frac{0 - V_{in}}{R_1} + \frac{0 - V_{out}}{R_2} = 0$$

$$\text{Voltage Gain}(A_v) = \frac{V_{out}}{V_{in}} = -\frac{R_2}{R_1}$$



Non-Inverting Operational Amplifier

When the signal is applied at the non-inverting input, the resulting circuit is known as Non-Inverting Op-Amp. In this amplifier the output is exactly in phase with the input i.e. when a positive voltage is applied to the circuit, the output will also be positive. By assuming the Op-Amp is ideal, then concept of virtual short can be applied i.e. the voltage at the inverting and non-inverting terminal is equal.



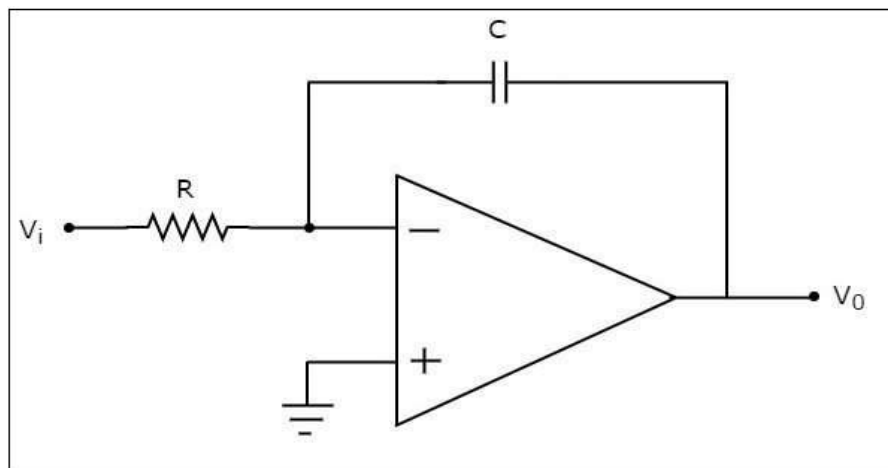
Applying KCL at the inverting node

$$\frac{V_{in} - V_{out}}{R_2} + \frac{V_0 - 0}{R_1} = 0$$

$$\text{Voltage Gain}(A_v) = \frac{V_{out}}{V_{in}} = 1 + \frac{R_2}{R_1}$$

Integrators

An integrator is an electronic circuit that produces an output that is the integration of the applied input. An op-amp based integrator produces an output, which is an integral of the input voltage applied to its inverting terminal. The circuit diagram of an op-amp based integrator is shown in the following figure –



In the circuit shown above, the non-inverting input terminal of the op-amp is connected to ground. That means zero volts is applied to its non-inverting input terminal.

According to virtual short concept, the voltage at the inverting input terminal of op-amp will be equal to the voltage present at its non-inverting input terminal. So, the voltage at the inverting input terminal of op-amp will be zero volts.

The nodal equation at the inverting input terminal is –

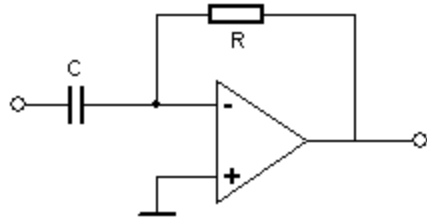
$$\begin{aligned}\frac{0 - V_i}{R} + C \frac{d(0 - V_0)}{dt} &= 0 \\ \Rightarrow \frac{-V_i}{R} &= C \frac{dV_0}{dt} \\ \Rightarrow \frac{dV_0}{dt} &= -\frac{V_i}{RC} \\ \Rightarrow dV_0 &= \left(-\frac{V_i}{RC}\right) dt\end{aligned}$$

Integrating both sides of the equation shown above, we get –

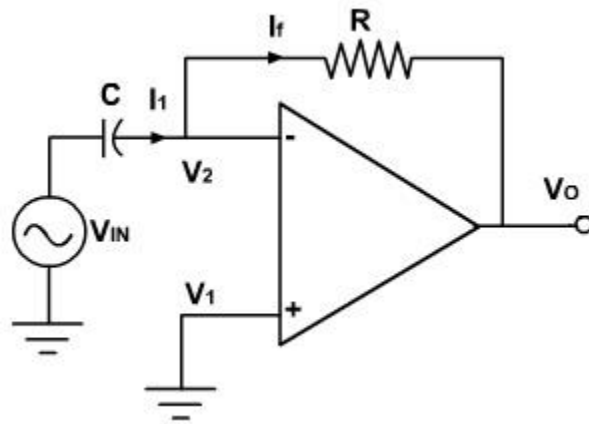
$$\begin{aligned}\int dV_0 &= \int \left(-\frac{V_i}{RC}\right) dt \\ \Rightarrow V_0 &= -\frac{1}{RC} \int V_i dt\end{aligned}$$

Differentiators

In an op-amp differentiator circuit, the output voltage is directly proportional to the input voltage rate of change with respect to time, which means that a quick change of the input voltage signal, then the high o/p voltage will change in response. As the output of an op-amp differentiator circuit is proportional to the change in input. When the inputs of the differentiator circuit are standard waveforms like sine, square, triangular then the output waveforms will be very different.



The operational amplifier circuit generates an output voltage which is proportional to the time derivative input voltage. So, this op-amp circuit is called as the differentiator. Assume the ground terminal which is denoted with G in the above circuit, where the flow of current through the ground terminal is equivalent to the flow of current out, we can write is as



In the above circuit, the op-amp node voltage at inverting terminal is zero then the flow of current through the capacitor can be written as

$$I_{in} = I_f$$

$$\text{Where } I_f = -V_{out}/R_f$$

The capacitor charge equals the voltage with capacitance times across the capacitor

$$Q = C \times V_{in}$$

Therefore the charge rate change is

$$dQ/dt = C dV_{in}/dt$$

But the dQ/dt is the current through the capacitor

$$I_{in} = C dV_{in}/dt = I_f$$

$$-V_{out}/R_f = C dV_{in}/dt$$

An ideal output voltage (V_{out}) for the operational amplifier differentiator is written as

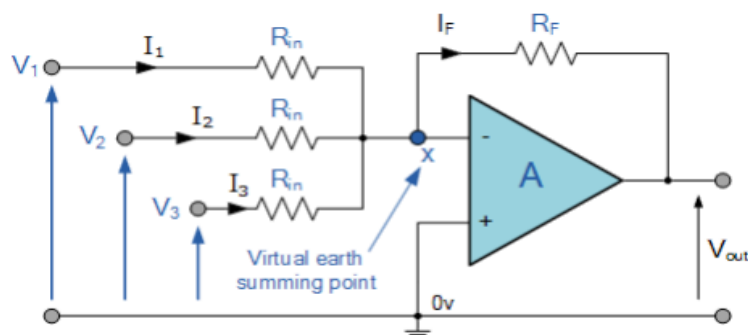
$$V_{out} = - R_f C dV_{in}/dt$$

Thus, the output voltage is a constant input voltage derivative $-R_f C$ times of the input V_{in} voltage with respect to time. Here sign minus $(-)$ specifies the phase shift (180°) as the input signal is given to the input inverting terminal of the op-amp.

Summing amplifiers

The Summing Amplifier is another type of operational amplifier circuit configuration that is used to combine the voltages present on two or more inputs into a single output voltage.

Summing Amplifier Circuit



In this simple summing amplifier circuit, the output voltage, (V_{out}) now becomes proportional to the sum of the input voltages, V_1 , V_2 , V_3 , etc. Then we can modify the original equation for the inverting amplifier to take account of these new inputs thus:

$$I_F = I_1 + I_2 + I_3 = - \left[\frac{V_1}{R_{in}} + \frac{V_2}{R_{in}} + \frac{V_3}{R_{in}} \right]$$

$$\text{Inverting Equation: } V_{out} = - \frac{R_f}{R_{in}} \times V_{in}$$

$$\text{then, } -V_{out} = \left[\frac{R_F}{R_{in}} V_1 + \frac{R_F}{R_{in}} V_2 + \frac{R_F}{R_{in}} V_3 \right]$$

Summing Amplifier Equation

$$-V_{out} = \frac{R_F}{R_{IN}} \left(V_1 + V_2 + V_3 \dots \text{etc} \right)$$